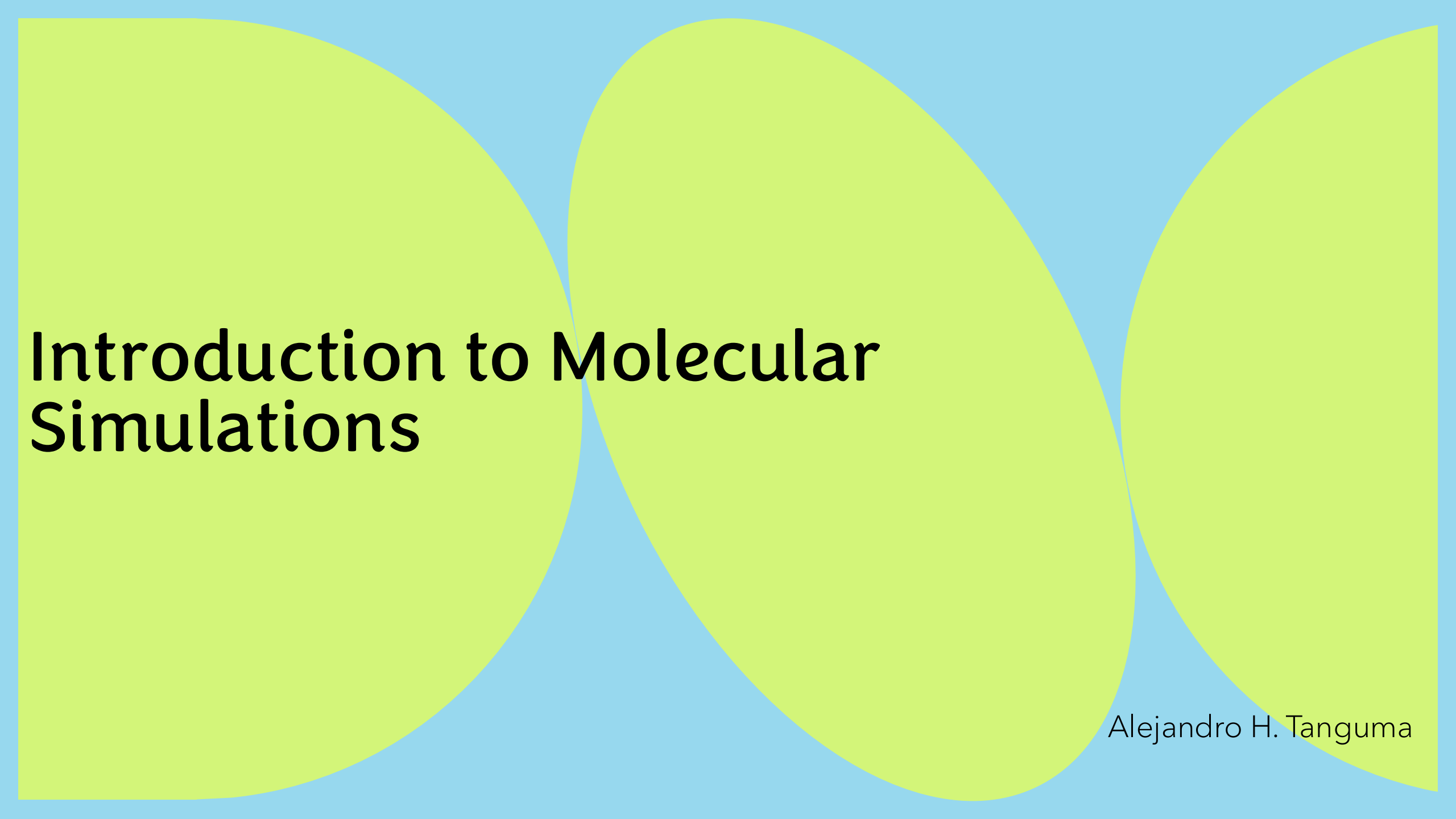




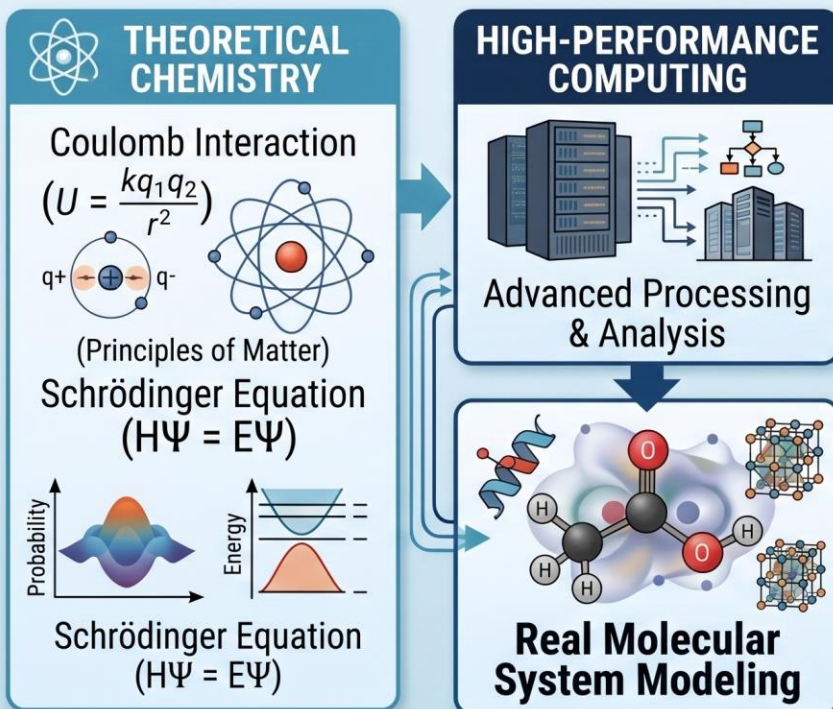
Introduction to Molecular Simulations

Alejandro H. Tanguma

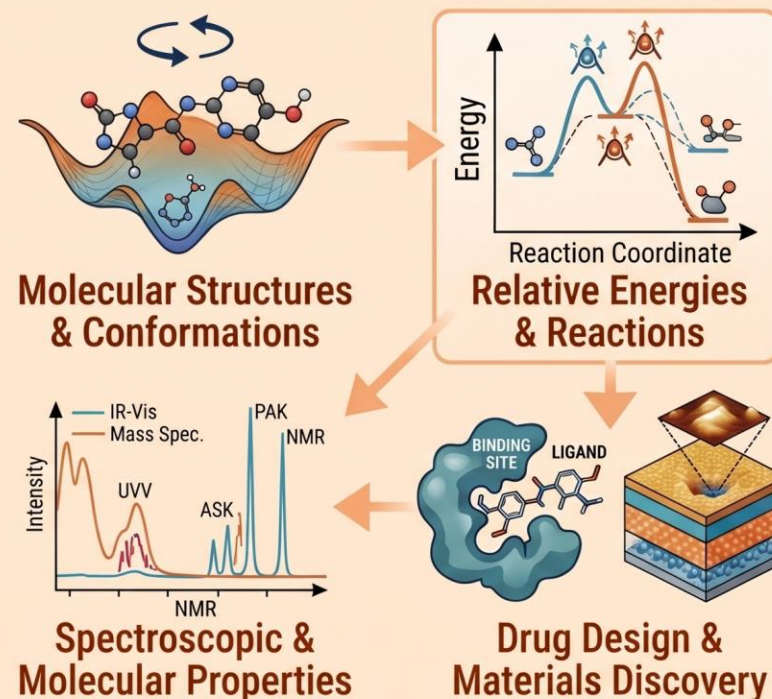
What is computational chemistry

WHAT IS COMPUTATIONAL CHEMISTRY & WHAT IS IT USED FOR?

WHAT IT IS

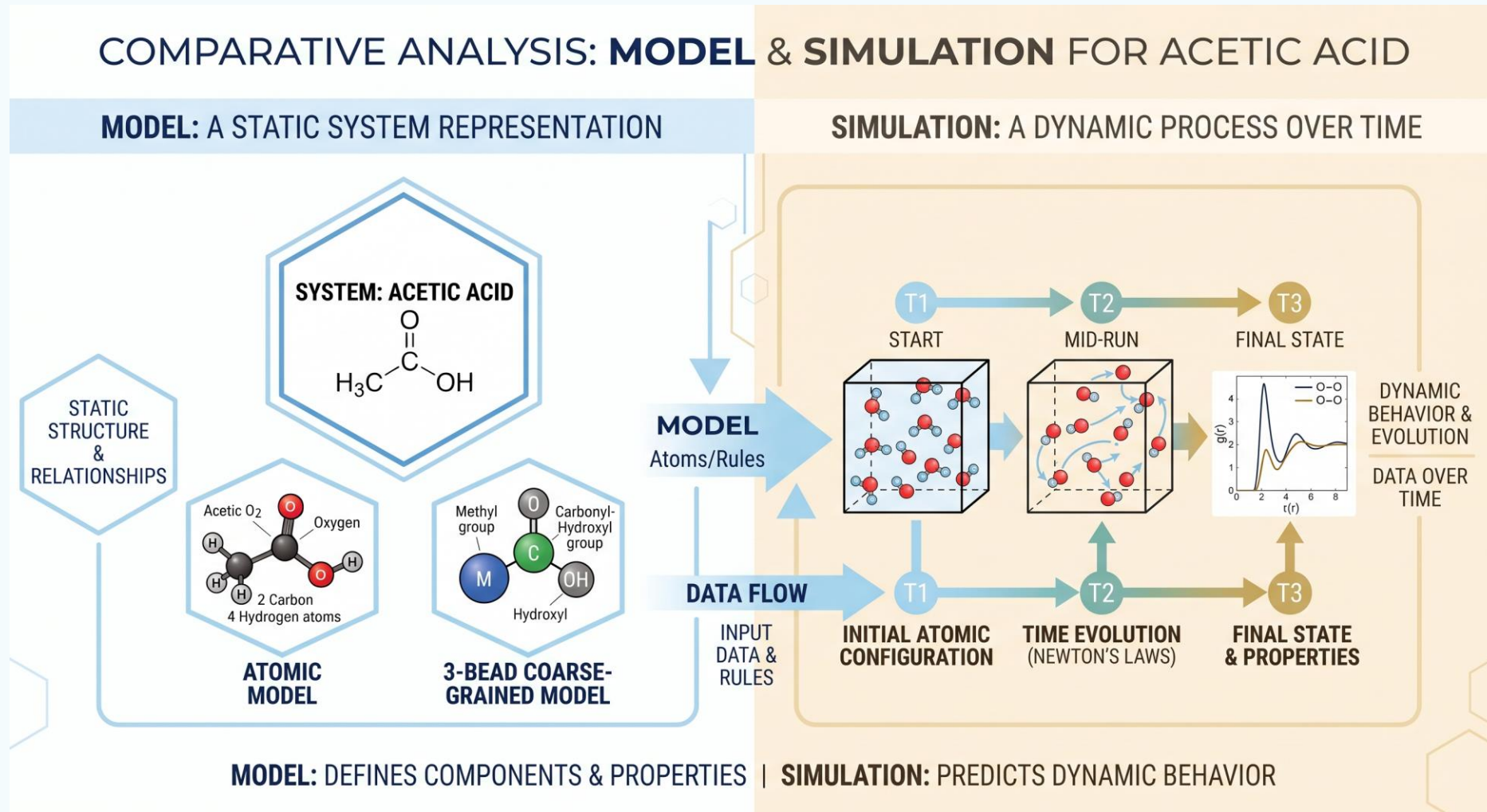


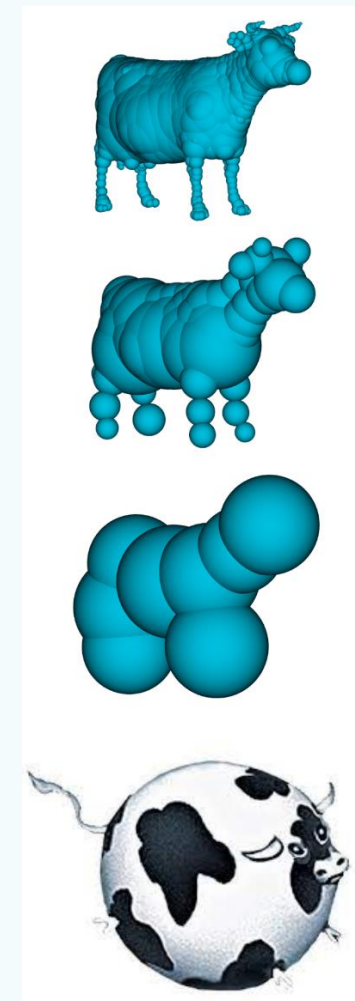
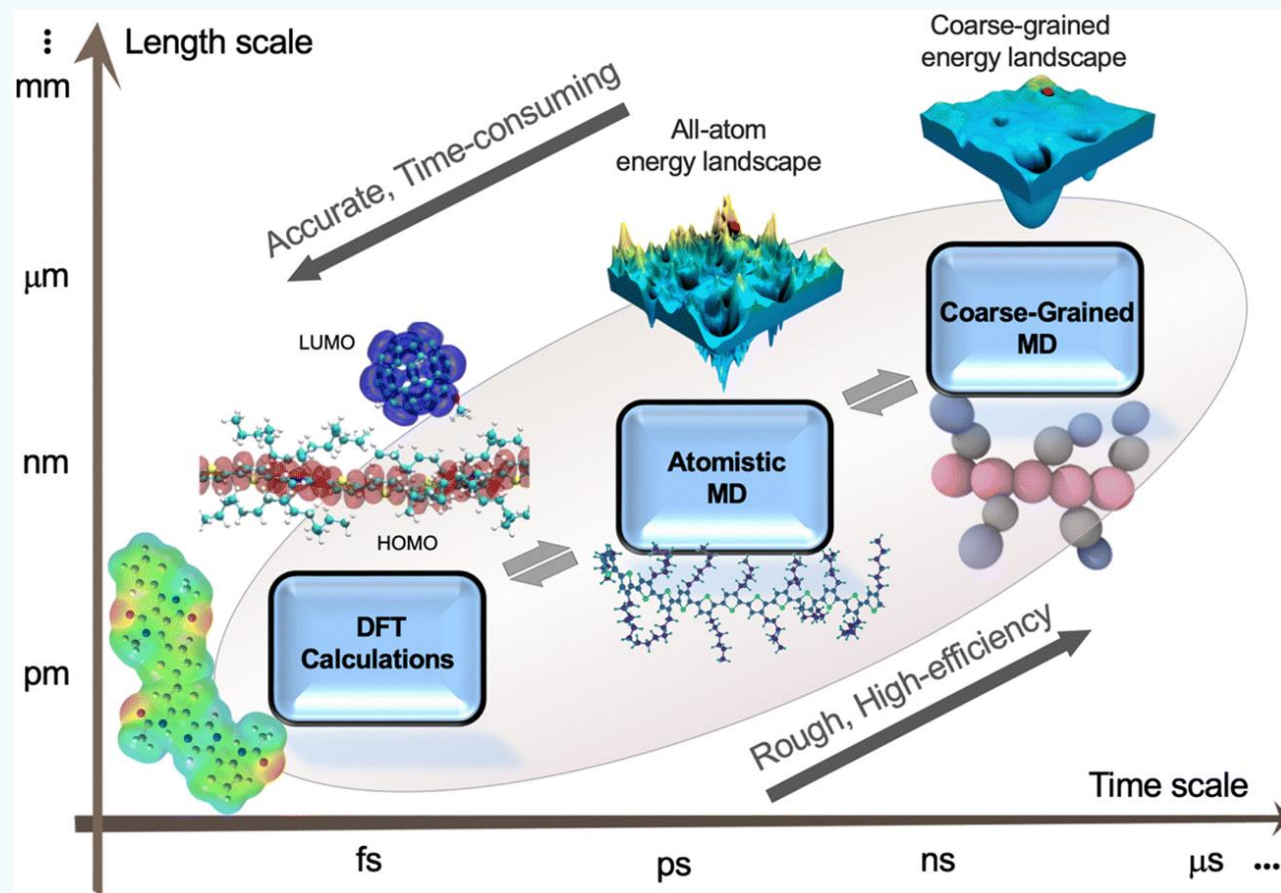
APPLICATIONS & PREDICTIONS



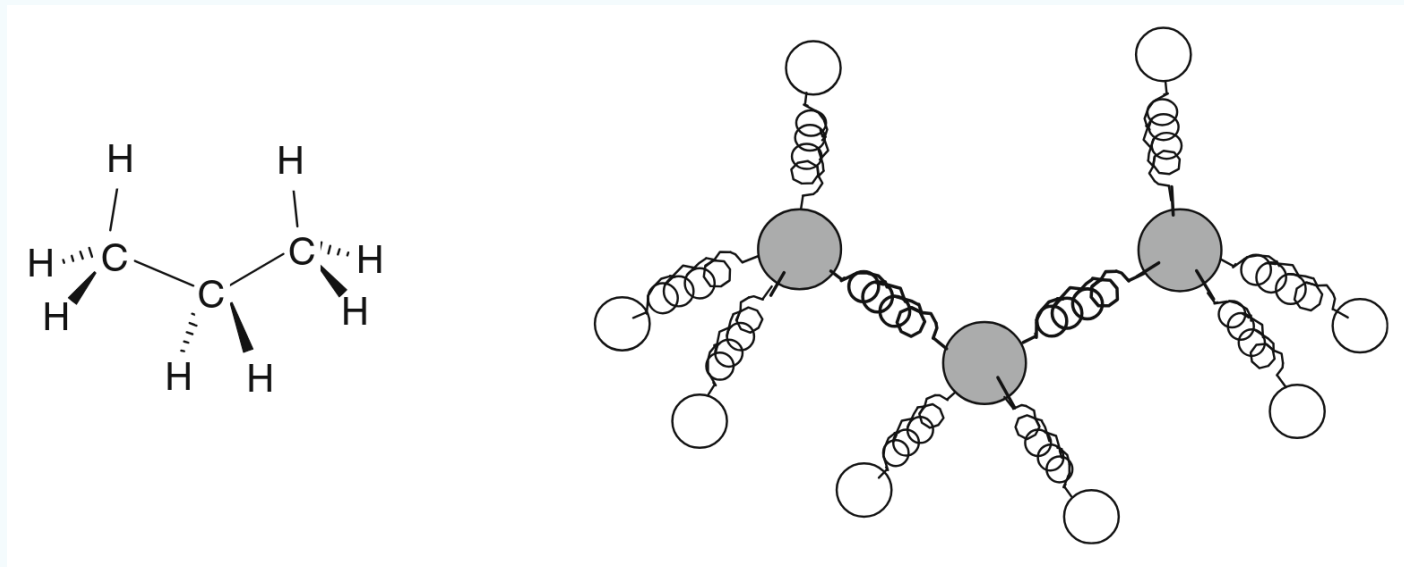
Bridge Between Theory & Experiment: Simulating Real Molecular Behavior

Models and simulations





Molecular mechanics



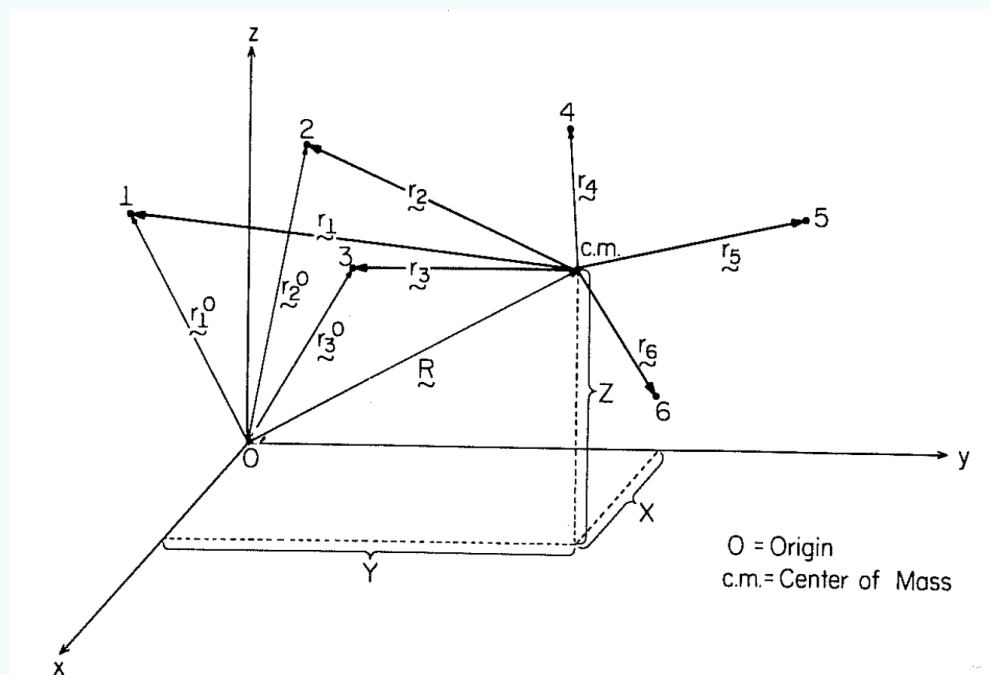


Figure 2-1 Cartesian coordinate axis system. O is the origin of the coordinate system chosen by convenience to be the laboratory frame and c.m. is the center of mass based in the many-particle frame. The r_i^0 and r_i vectors originate at the arbitrary origin and c.m., respectively, and terminate at the point masses designated by 1, 2, $\mathbf{R}$ is the vector from the arbitrary origin to the c.m. and X , Y , and Z are the Cartesian components of $\mathbf{R}$.

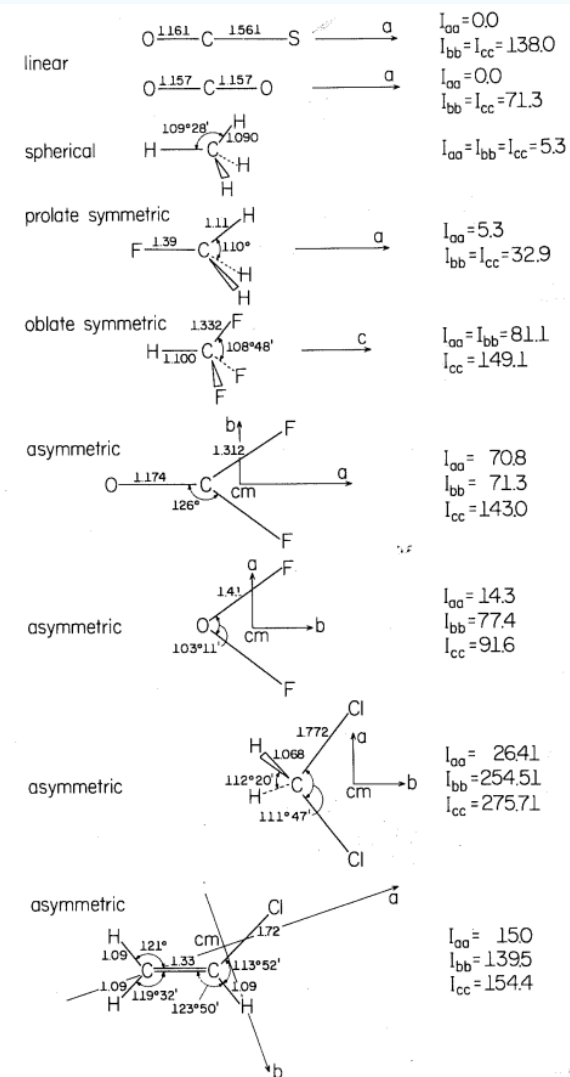
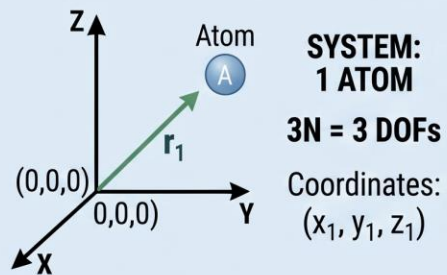


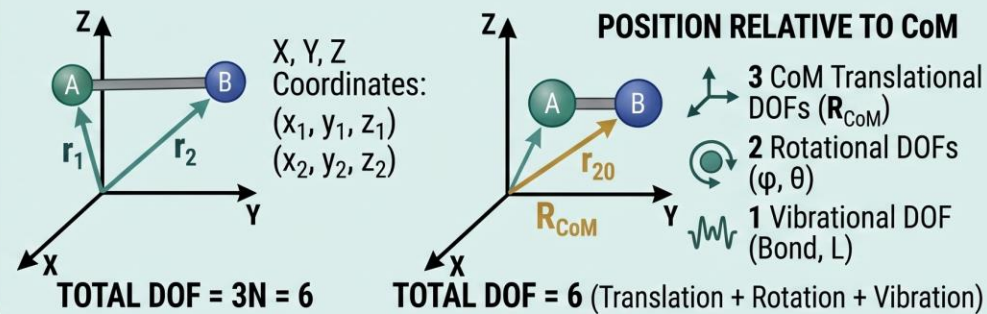
Figure 2-2 Moments of inertia of several molecules in the principal inertial axis system (a , b , c) for the common isotopes (^{16}O , ^{12}C , ^{32}S , ^{19}F , and ^{35}Cl). The bond distances are in units of $\text{\AA} = 10^{-8} \text{ cm}$. The units of the moments are $10^{-40} \text{ g} \cdot \text{cm}^2$, or $10^{-47} \text{ kg} \cdot \text{m}^2$.

DEGREES OF FREEDOM (DOF) & PARTICLE COORDINATES

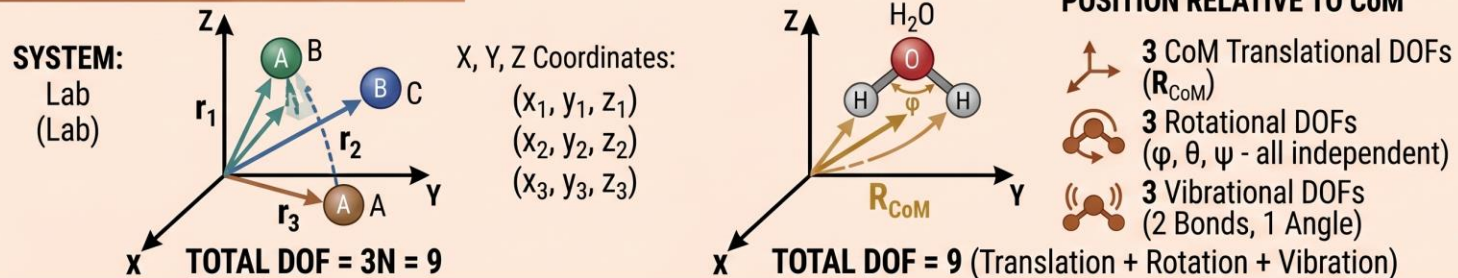
1 PARTICLE SYSTEM (N=1)



2 PARTICLE SYSTEM (N=2)



3 PARTICLE SYSTEM (N=3)

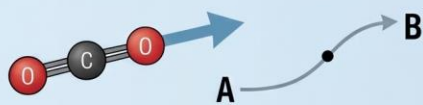


MOLECULAR MOTION TYPES & TYPICAL TIMESCALES (Small Molecules)

TYPES OF MOLECULAR MOTION

TRANSLATION

Movement of Center of Mass (CoM)



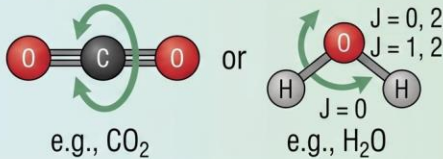
e.g., CO₂, of Center of Mass (CoM)

TRANSLATION

Molecule travels from point A to point B.

ROTATION

Rotation around Center of Mass



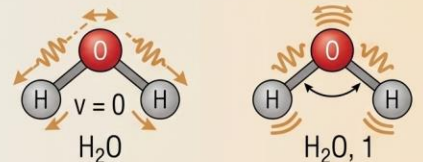
e.g., CO₂

J = 0, 2
J = 1, 2
J = 0
e.g., H₂O

ROTATION

Molecule spins without displacing.

VIBRATION



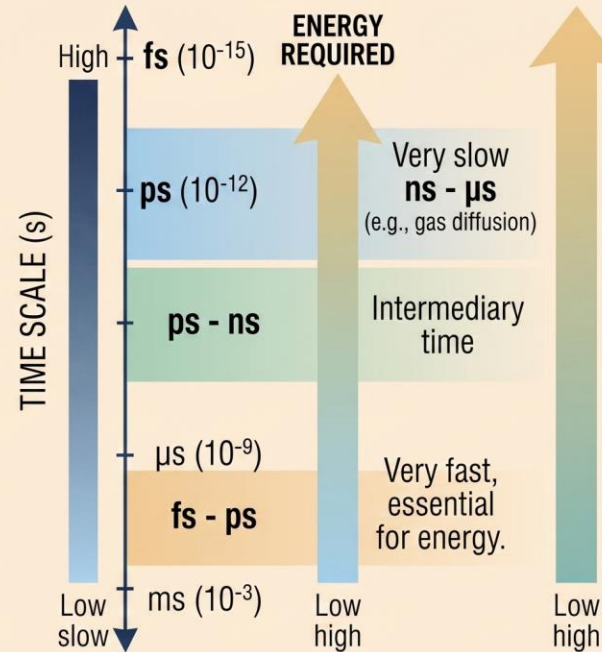
v = 0
H₂O

H₂O, 1

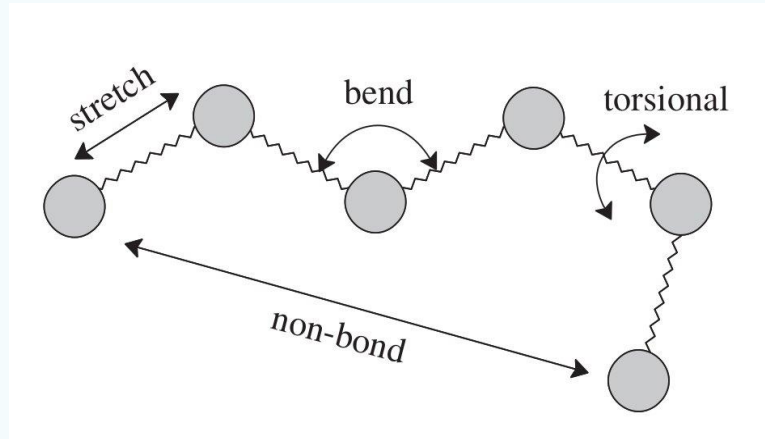
Vibrational Modes: Stretches & Bends

Atoms oscillate around their equilibrium positions.

TYPICAL TIMESCALES



SUMMARY: TRANSLATION: Macro displacement | ROTATION: Fast spinning | VIBRATION: Ultrafast oscillation



$$E = \sum_{bonds} E_{stretch} + \sum_{angles} E_{bend} + \sum_{dihedrals} E_{torsion} + \sum_{pairs} E_{nonbond}$$

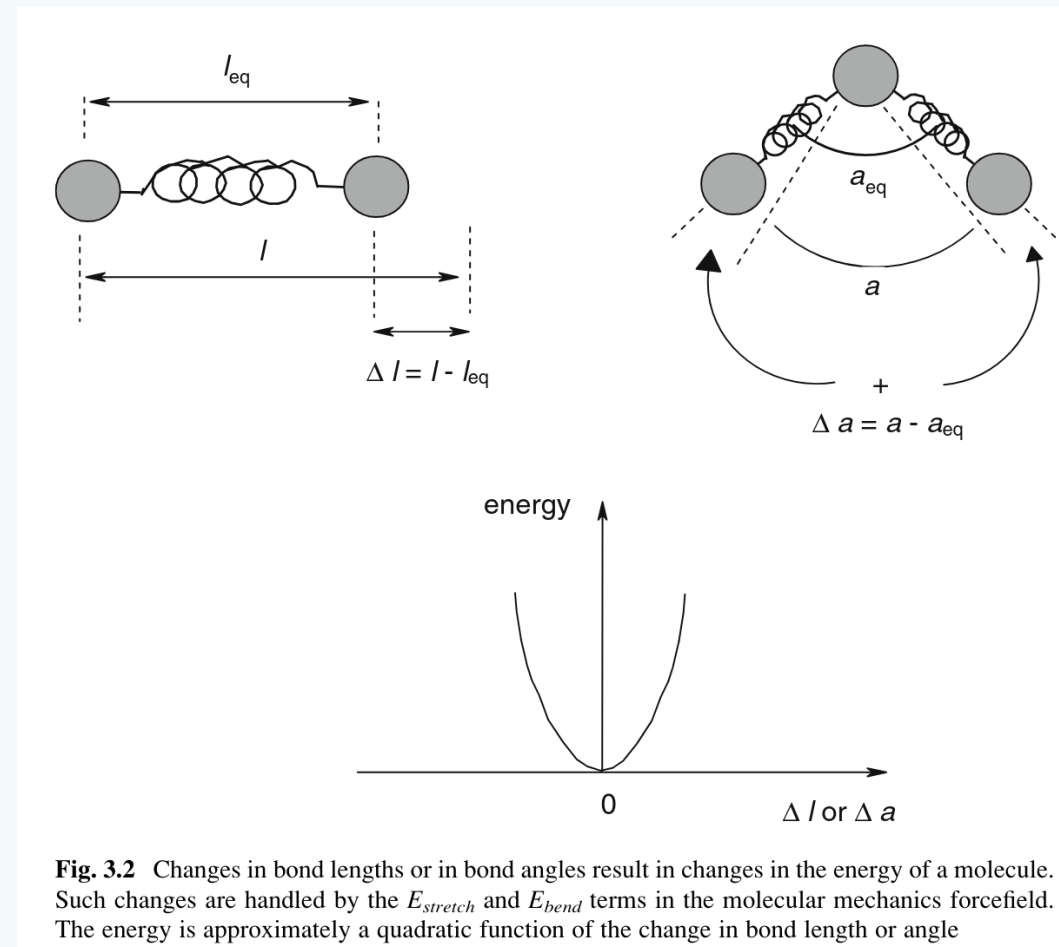
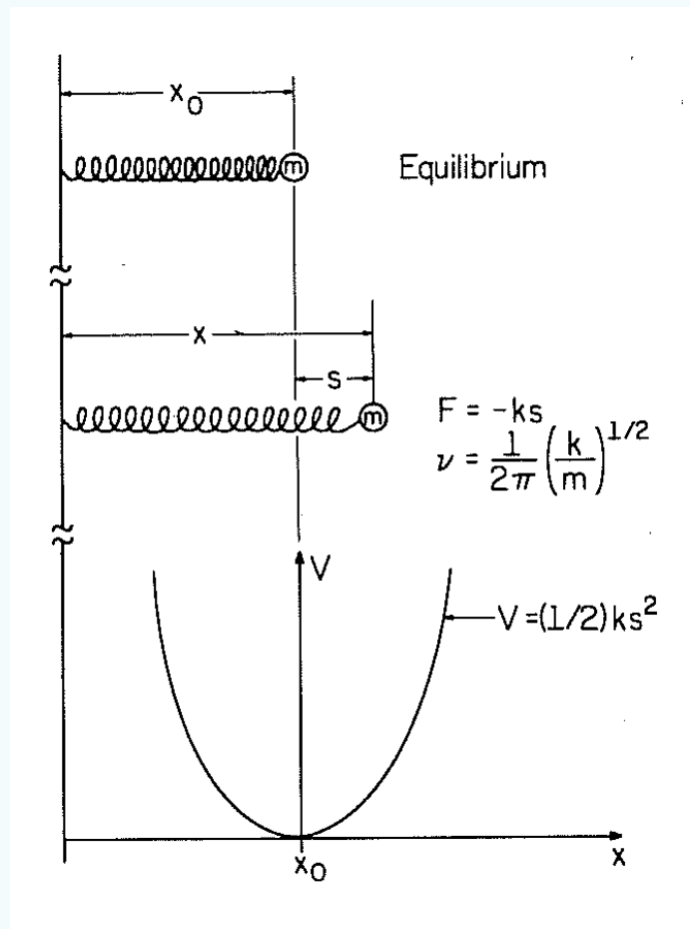
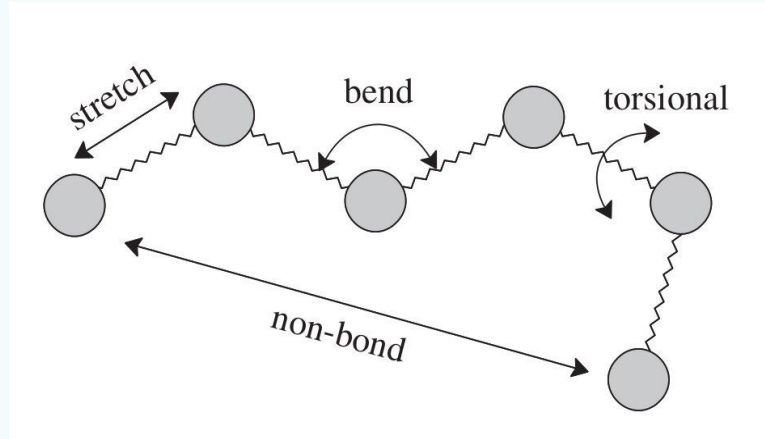


Fig. 3.2 Changes in bond lengths or in bond angles result in changes in the energy of a molecule. Such changes are handled by the $E_{stretch}$ and E_{bend} terms in the molecular mechanics forcefield. The energy is approximately a quadratic function of the change in bond length or angle



$$\begin{aligned}
 \mathcal{V} = & \sum_{\text{bonds}} \frac{1}{2} k_r (r_{ij} - r_0)^2 + \sum_{\text{angles}} \frac{1}{2} k_\theta (\theta_{ijk} - \theta_0)^2 \\
 & + \sum_{\text{torsions}} \sum_n k_{\phi,n} [\cos(n\phi_{ijk\ell} + \delta_n) + 1] + \sum_{\text{non-bonded pairs}} \left[\frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} + \frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} \right].
 \end{aligned}$$

https://github.com/TangumaAH/intro_MD